Technical Document #386: Data Engineering - Comprehensive Overview

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Data warehouses store structured data optimized for analytical queries. They use columnar storage and support complex aggregations.

Data quality ensures data is accurate, complete, and consistent. Data validation, cleansing, and monitoring are essential components.

Data governance establishes policies and procedures for managing data assets. It covers data privacy, security, and compliance.

Data lakes store raw data in its native format. They support structured, semi-structured, and unstructured data at any scale.

Natural language processing has made significant advances with large language models. These models can understand and generate human-like text with remarkable fluency.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Edge computing processes data closer to its source, reducing latency and bandwidth usage. It is essential for real-time applications like autonomous vehicles.

Key Takeaways:

- ETL (Extract, Transform, Load) is a process that extracts data from source systems, transforms it to fit business rules, and loa

- Partitioning divides data into smaller, more manageable pieces.

- Data pipelines automate the movement and transformation of data between systems.